



R1.1 Measuring enthalpy changes

Reactivity 1 · What drives chemical reactions? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Enthalpy (H) is the heat content of a system at constant pressure; an exothermic reaction releases heat to the surroundings (ΔH negative, temperature of surroundings rises) while an endothermic reaction absorbs heat (ΔH positive, temperature falls). Enthalpy changes can be measured experimentally using calorimetry, applying $q = mc\Delta T$, where q is the heat transferred to/from the surroundings (usually a known mass of water/solution), m is the mass of that surroundings, c is its specific heat capacity, and ΔT is the temperature change; this heat is then converted to a molar enthalpy change by dividing by the amount (mol) of the limiting reagent. Standard enthalpy changes (ΔH°) are measured under standard conditions (100 kPa, usually 298 K, standard states), and Hess's law states that the overall enthalpy change for a reaction is independent of the pathway taken, since enthalpy is a state function.

Worked example: follow the reasoning

50.0 cm³ of 1.00 mol dm⁻³ HCl(aq) is mixed with 50.0 cm³ of 1.00 mol dm⁻³ NaOH(aq), both initially at 20.0 °C. The temperature rises to 26.8 °C. Calculate the enthalpy change of neutralisation per mole of water formed. (Assume the density of the solution is 1.00 g cm⁻³ and its specific heat capacity is 4.18 J g⁻¹ K⁻¹.)

Step 1: Total mass of solution = 50.0 + 50.0 = 100.0 g.

Step 2: $\Delta T = 26.8 - 20.0 = 6.8$ °C (= 6.8 K).

Step 3: $q = mc\Delta T = 100.0 \times 4.18 \times 6.8 = 2842$ J = 2.842 kJ (heat released to the solution, so the reaction is exothermic).

Step 4: $n(\text{HCl}) = n(\text{NaOH}) = 0.0500 \text{ dm}^3 \times 1.00 \text{ mol dm}^{-3} = 0.0500 \text{ mol}$ (also = $n(\text{H}_2\text{O})$ formed, 1:1 stoichiometry).

Step 5: $\Delta H = -q / n = -2.842 / 0.0500 = -56.8$ kJ mol⁻¹.

Answer: $\Delta H(\text{neutralisation}) = -56.8$ kJ mol⁻¹.

Quick check · multiple choice

In a calorimetry experiment, the temperature of the surroundings decreases during a reaction. What does this indicate?

- A The reaction is exothermic and ΔH is negative
- B The reaction is endothermic and ΔH is positive
- C The reaction has reached equilibrium
- D The reaction has zero enthalpy change

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student adds 4.20 g of solid ammonium nitrate, NH_4NO_3 ($M = 80.0 \text{ g mol}^{-1}$), to 100.0 cm³ of water in an insulated cup, initially at 22.5 °C. The salt dissolves and the temperature falls to 18.7 °C. [7]
- (a) Calculate the amount, in mol, of NH_4NO_3 used. [1]
 - (b) Calculate the heat absorbed during dissolution, assuming the density of the solution is 1.00 g cm⁻³ and its specific heat capacity is 4.18 J g⁻¹ K⁻¹. State any assumption made about the mass used in this calculation. [3]
 - (c) Calculate the molar enthalpy change of solution, ΔH , for NH_4NO_3 , including its sign. [2]
 - (d) Suggest one modification to the experimental procedure that would improve the accuracy of the measured ΔH value. [1]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $n = 4.20 / 80.0 = 0.0525 \text{ mol}$. [1]

(b) $\Delta T = 22.5 - 18.7 = 3.8 \text{ }^{\circ}\text{C (K)}$. Assumption: mass of solution $\approx$ mass of water used = 100.0 g (the small added mass of solid is often approximated as not changing the total significantly, or total mass = 104.2 g if included). Using $m = 100.0 \text{ g}$: $q = mc\Delta T = 100.0 \times 4.18 \times 3.8 = 1588 \text{ J} \approx 1.59 \text{ kJ}$. [3]

(c) The temperature fell, so the process absorbed heat from the solution (endothermic): $\Delta H = +q/n = +1.59 / 0.0525 = +30.3 \text{ kJ mol}^{-1}$. [2]

(d) Any valid answer, e.g. use a lid/better insulation on the cup to reduce heat loss/gain to the surroundings, or use a more precise thermometer/temperature sensor with smaller graduations, or stir continuously to ensure uniform temperature. [1]

Quick check answer

B. A temperature decrease means the reaction absorbed heat from the surroundings, so it is endothermic and ΔH is positive.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.